

ENVIRONMENTAL REVIEW & WASTE MANAGEMENT INSTRUCTIONS FOR: EXCAVATION PERMITS

PERMIT #:	LCAAP AREA #:	Disturbed Soil Area of Excavation: <i>Areas ≥ 1 acre require MDNR Land Disturbance Permit</i>	
WITHIN AREA OF CONCERN (AOC): ☐ Yes ☐ No <i>If Yes, Site visit required.</i> Site Visit Date:			
CONTAMINANT OF CONCERN (COC)	MEDIA OF CONCERN:		
	SOIL	GROUNDWATER	CONCRETE/ASPHALT
SEMI -VOLATILE ORGANIC COMPOUNDS (SVOC)	☐	☐	☐
VOLATILE ORGANIC COMPOUNDS (VOC)	☐	☐	☐
POLYCHLORINATED BIPHENYL (PCB)	☐	☐	☐
POLYCYCLIC AROMATIC HYDROCARBONS (PAH)	☐	☐	☐
METAL	☐	☐	☐
EXPLOSIVES	☐	☐	☐
MUNITIONS & EXPLOSIVES OF CONCERN (MEC)	☐	☐	☐
RADIONUCLIDES	☐	☐	☐

SPECIAL CONDITIONS OF WASTE MANAGEMENT, RECYCLING, AND/OR DISPOSAL:

Minimum Suggested Personal Protective Equipment (PPE) – Required in addition to requirements on LC-5145 (if in direct contact with ground water):

☒ Safety Goggles ☒ Disposable Gloves ☐ Respiratory Protection ☐ Chemical Resistant Clothing

***Proceeding information is required to be reviewed by the contractor/subcontractor's Health and Safety Officer. Appropriate PPE must be used for the chemicals of concern in the soil and groundwater.**

General Instructions:

- Operating Contractor Engineer and Individuals performing excavation must review attached:
 - Groundwater Depth Information for the designated LCAAP Area
 - IWOU – Installation Wide Land Use Controls Information
 - Any additional Area Land Use Control Information
 - Contaminants of Concern Information
 - Site Maps
- Contact Operating Contractor Environmental Engineering if any off color or strange smelling soil or water is encountered.
- Use the following information to determine how to manage any media of concern:
 - Hydro Excavation Debris**
 - Take to on-site drying bed. Contact Project Engineer for guidance on what drying bed will be utilized for the project. All efforts will be made to keep hydro excavation debris segregated by project location to allow the dried debris to be returned to the excavation area in accordance with F.i. of this procedure.**
 - If unable to keep segregated by project and/or return to project area the excavated debris will either be disposed of off-site in accordance with F.ii. or F. iii of this procedure.**
 - GROUNDWATER:** Containerize all groundwater and dispose of at 93CC, unless directed otherwise due to specific area restrictions (see Special Conditions above).

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C. CONCRETE TRUCK WASHOUT: Utilize Best Practices

- i. Return excess concrete with truck.
- ii. Rinse chute & tools into equipment bucket or plastic lined skid box (dispose in dumpster after project is complete.)
- iii. Create designated lined washout at project location for larger jobs. Washout area must be removed from LCAAP at the completion of the project.

D. STORM WATER:

- i. Contractor shall apply for and receive a Land Disturbance Permit through the Missouri Department of Natural Resources, prior to any excavation activity, for projects expected to disturb vegetative cover or expose soil in quantities 1 acre or greater.
- ii. Erosion control measures shall be implemented on disturbed areas according to Best Management Practices, especially on larger disturbed areas and in areas where there is a high potential for erosion due to location and grade.
- iii. Soil awaiting sampling results must be placed on plastic sheeting, and covered with plastic/tarp.

E. CLEAN FILL (ASPHALT & CONCRETE): Clean fill is “uncontaminated soil, rock, sand, gravel, concrete, asphaltic concrete, cinder blocks, brick, minimal amounts of wood and metal and inert (nonreactive) solids... for fill, reclamation or other beneficial use,” according to Section 206.200(4), RSMo. Minimal means the smallest amount possible. For example, concrete containing wire mesh or reinforcing rods (rebar) may be used for clean fill if you removed the exposed rebar before use. Although clean fill is not regulated as solid waste, some recovered materials, such as asphaltic concrete, may be used only in locations where they will not contact groundwater or surface water. Under no circumstances are shingles, sheet rock or wood wastes defined as clean fill.

- i. Waste asphalt and/or concrete materials are to be disposed of at an approved landfill (Pink Hill, Courtney Ridge, or Johnson County).
- ii. NO DUMPING of concrete and/or asphalt anywhere on plant is permitted without prior approval.
- iii. All waste asphalt and/or concrete to be disposed of should be free of:
 1. Oil & Grease
 2. Explosive Residues
 3. Chemical Contamination
 4. **NOTE: If contamination is seen or suspected contact Operating Contractor Environmental Engineering at 816-656-1719 or 816-656-1696 for an inspection.**
- iv. DISPOSAL ALTERNATIVES
 1. Disposal alternatives for concrete in excess of 100 cubic yards will be considered on a case-by-case basis. Follow and submit sampling plan process for clean fill as described below.

F. SOIL MANAGEMENT: Due to LCAAP Land Use Control restrictions: Under no circumstances will soil from LCAAP be allowed for reuse off site as clean fill or otherwise.

- i. Reincorporate excavated soil within 50-feet of excavated area footprint.
 1. If the soil is discolored, has a distinct odor, etc. the contractor should stop work and contact Operating Contractor Environmental Engineering for guidance.
 2. If Environmental determines that it is acceptable for the soil to be used within the original location, and not moved elsewhere; Smart map shall be updated to reflect observed conditions.
- ii. Soil that will not be incorporated within 50 feet of the excavation area footprint must be sampled and disposed off-site. Disposal off-site to approved landfill (Courtney Ridge, Johnson County) is the only acceptable method of disposal/removal from LCAAP.
 1. Sampling Requirements: The following analysis must be run and reviewed by Environmental Engineering prior to removing soil from the site
 - a. *Operating Contractor projects*
 - i. One (1) small scoop will be taken from each excavator bucket load excavated, each small scoop is to be placed in a sample bucket provided by LCAAP Environmental Engineering, to create the composite sample

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- ii. A minimum of one (1) composite sample is required for every project if <100 cubic yard (cy) or for every 100 cy of excavated soil/clean fill thereafter
 - iii. The sample will be provided to Operating Contractor Environmental Engineering to be sent off-site for analysis
 - b. *All other projects*
 - i. One (1) small scoop will be taken from each excavator bucket load excavated, each small scoop is to be placed in a sample bucket, to create the composite sample
 - ii. A minimum of one (1) composite sample is required for every project if <100 cubic yard (cy) or for every 100 cy of excavated soil/clean fill thereafter
 - iii. The sample will be provided to an approved third party laboratory by contractor for analysis and results presented to LCAAP USG Environmental.
 - c. Samples will be analyzed by TCLP for RCRA Metals, Volatiles, Semi-Volatiles, Pesticides, and Explosives (if applicable). Analytical results may take between 3-7 business days, plan for soil storage location and project time.
 - d. Soil awaiting sampling results must be placed in a lined roll-off container (recommended) or on plastic sheeting, and covered with plastic/tarp.
 - e. Disposition determined by sampling results.
 - i. Analytical below, at, or above LCAAP background levels (see Section E. iii. 2.d. Table 13), but below hazardous waste upper tolerance limits (UTL) will be disposed to landfill as a special waste.
 - ii. Analytical above LCAAP background and above any of the hazardous waste UTL will be disposed as a Hazardous Waste at an approved Hazardous Waste landfill/treatment facility.
- iii. **DISPOSAL ALTERNATIVES:** Alternate on site soil/clean fill management may be available on a case-by-case basis; requires a written plan and Operating Contractor environmental and USG environmental approval. *Only uncontaminated soil/clean fill will be approved for reuse on plant outside of the original excavation footprint.*
- 1. Reuse of Soil and Clean Fill onsite
 - a. Contractor plan shall include at a minimum:
 - i. Review of 1 acre Land Use Disturbance Permit requirement applicability
 - ii. Determination if additional number of samples/analysis (outside of section 3. E. iii. 2.) are necessary
 - iii. Quantities of soil/clean fill to be moved and left in place
 - iv. Map of locations of removal and reuse site(s)
 - v. Timeline of work to be performed – Soil/clean fill shall not be left in piles after the completion of the project. All soil/clean fill must be reused or disposed during the course of the project.
 - b. Submit plan to both Operating Contractor Environmental Engineering and LCAAP USG Environmental for consideration and approval. LC-4367E **must be signed** prior to reuse being approved.
 - 2. Sampling Requirements: The following frequency/analytes at a minimum must be run and presented to Operating Contractor Environmental Engineering prior to soil/clean fill being reused onsite. If below sampling is not feasible, submit sampling frequency and analytes in plan.
 - a. *Operating Contractor projects*
 - i. One (1) small scoop will be taken from each excavator bucket load excavated to create the composite sample
 - ii. A minimum of one (1) composite sample is required for every project if <40 cubic yard (cy) or for every 40 cy of excavated soil/clean fill thereafter
 - iii. The sample will be provided to Operating Contractor Environmental Engineering to be sent off-site for analysis
 - b. *All other projects*
 - i. One (1) small scoop will be taken from each excavator bucket load excavated to create the composite sample
 - ii. A minimum of one (1) composite sample is required for every project if <40 cubic yard (cy) or for every 40 cy of excavated soil/clean fill thereafter
 - iii. The sample will be provided to an approved third party laboratory by contractor for analysis and results presented to LCAAP USG Environmental.
 - c. Total analysis for the following constituents:

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Aluminum	Cadmium	Manganese	Strontium	Explosives
Antimony	Chromium	Mercury	Thallium	Pesticides
Arsenic	Copper	Nickel	Zinc	PCBs
Barium	Iron	Selenium	Volatiles	
Beryllium	Lead	Silver	Semi-volatiles	

- d. The results of the tests will be compared to background results (Table 13 below) and applicable EPA and State standards.

Table 13
Summary of Final Background UTLs for Soil and Groundwater
Lake City Army Ammunition Plant, Independence, Missouri

Metal	Soil UTL (mg/kg)	Groundwater UTL (ug/l)
Aluminum	--	3,600
Antimony	4.2	0.42
Arsenic	12	5.3
Barium	570	630
Beryllium	2	--
Cadmium	3.9	0.43
Chromium	53	22
Copper	43	9.2
Iron	24,000	30,000
Lead	39	17
Manganese	2,700	3,800
Mercury	0.16	--
Nickel	36	23
Selenium	--	18
Silver	3	--
Strontium	--	1,500
Thallium	--	0.36
Zinc	170	50

Notes:

- A background upper tolerance limit (UTL) has not been specified for the given medium.

G. Soil/Clean fill Brought Onsite

- i. Prior to placing soil/clean fill on plant the contractor must obtain recent third party analytical results (within 6 months of request) and submit the above Total analysis data (Section 3. E. iii. 2. c.) to the Operating Contractor Environmental Engineering department for review and approval.

Any questions or concerns are to be directed to Operating Contractor Environmental Engineering 816-656-1719 or 816-656-1696; US Government Environmental 816-796-7159 or your Facility Technical Representative.